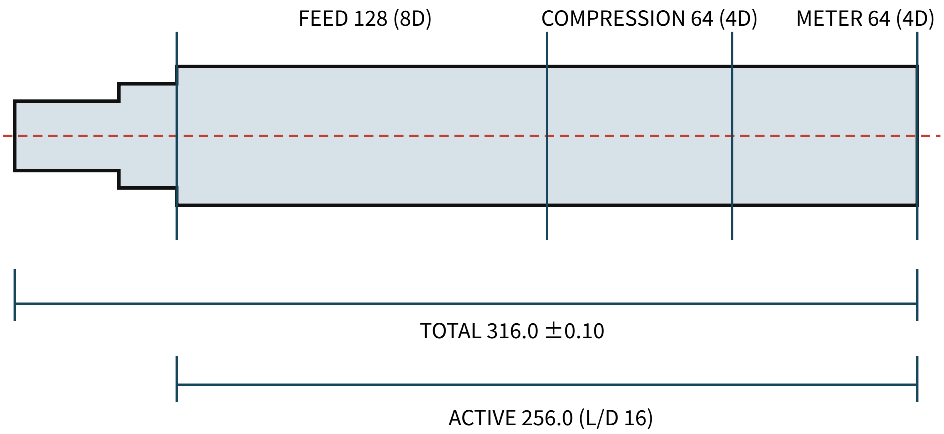
# 1. 16 mm × 16 L/D screw/barrel RFQ drawing

Revision: `solid-manifold-openmodelica-v0.4` / 단위: mm / 온도: 20 ±2 °C / 일반 모서리 C0.2–0.5, burr 없음

**FULL 발주 HOLD — EX-CPN-SCR/EX-CPN-BAR 공정 coupon과 공급사 DFM, Gate-3 승인 전 EX-SCR-01/EX-BAR-01 발주 금지**

## 1.1. EX-SCR-01

### EX-SCR-01 — 16 mm × 16D single screw RFQ drawing




Rear: Ø12 h6 x35, keyseat 4 P9 x2.5 deep · thrust journal Ø15 h6 x20 · neck 5

OD 15.92 -0.02/0 · pitch 16.00 ±0.03 · land 1.60 ±0.05 · single start RH

root Ø10.88 feed → linear Ø14.08 compression → Ø14.08 meter · end faces ⊥ A 0.03

Datum A: common journal axis · flight OD TIR ≤0.05/256 · drive-to-flight concentricity ≤0.03 · OD Ra≤0.8 µm

SCM440 QT 28–32 HRC → gas nitride 0.30–0.50 mm, surface 900–1100 HV · full part HOLD

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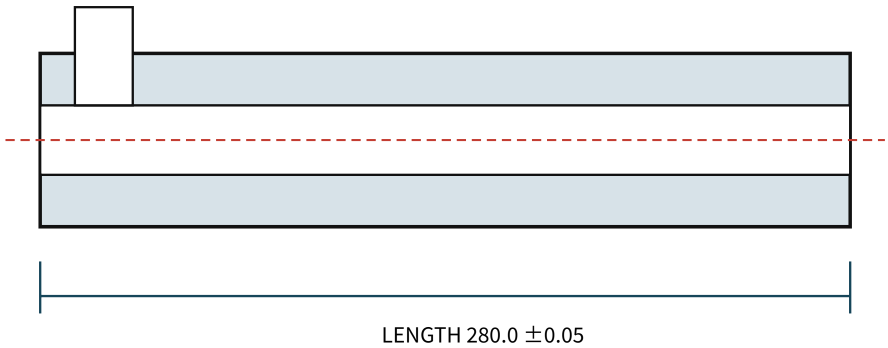
|---|---|
| 재료 | SCM440, normalized blank → rough turn → QT 28–32 HRC |
| 주요 치수 | Total 316.0 ±0.10; active 256.0; single-start RH; pitch 16.00 ±0.03; land 1.60 ±0.05; OD Ø15.92 −0.02/0 |
| Zone/root | Feed 128, root Ø10.88; compression 64, linear Ø10.88→Ø14.08; meter 64, root Ø14.08 |
| Journal | Drive Ø12 h6 ×35, KS/DIN key 4×4, keyseat 4 P9 wide ×2.5 +0.10/0 deep; thrust Ø15 h6 ×20; shoulder perpendicularity 0.03 to Datum A; flight start 0° ±5° from key plane |
| GD&T | Flight OD TIR ≤0.05/256; drive-to-flight concentricity ≤0.03; straightness ≤0.05/256 |
| 표면 | Flight OD Ra≤0.8 µm; root/flank Ra≤1.6 µm; weld repair 금지 |
| 열처리 | Gas nitride effective case 0.30–0.50, surface 900–1100 HV; journals mask; final OD grind between centres |

가공 route: normalized blank → datum centre drilling → rough turn → QT → finish journals/shoulders leaving flight allowance 0.15 → 4-axis flight mill → root/flank polish → gas nitride → flight OD grind → TIR/Ra/hardness 검사.

## 1.2. EX-BAR-01

### EX-BAR-01 — Ø34 / ID16.20 barrel RFQ drawing




OD Ø34.00 ±0.05 · bore Ø16.20 +0.02/0 after final hone · radial clearance 0.14–0.16

feed port 18 axial ×20, near edge 12.0 from rear Datum B · break edge R0.5

4x M5×0.8-6H depth10, PCD28 at 45° · bore straightness ≤0.05/256 · faces B/C ⊥ datum D axis 0.03

bore Ra 0.4–0.8 μm; SCM440 QT 28–32 HRC → gas nitride 0.30–0.50 mm, ≥900 HV

Assembly: B aligns screw active start; screw tip is 24.0 behind C. Final hone after nitride; report ID at B+20/140/260.

No weld/plating on bore · feed-port centre plane is angular datum · full part HOLD

| | |
|---|---|
| 재료 | SCM440 solid/seamless blank, QT 28–32 HRC |
| 주요 치수 | OD Ø34.00 ±0.05; L280.00 ±0.05; final bore Ø16.20 +0.02/0 |
| Port/thread | 18 axial ×20, rear edge 12.0 from Datum B, edge R0.5; front 4× M5×0.8 depth≥10 on PCD28 at 45/135/225/315° from port centre plane |
| GD&T | Bore straightness ≤0.05/256; bore-to-OD/register concentricity ≤0.05; end face perpendicularity 0.03 |
| 표면 | Final bore Ra 0.4–0.8 μm; no weld/plating in bore |
| 열처리 | Gas nitride 0.30–0.50, ≥900 HV; final hone 후 effective case ≥0.25 |
| 검사 | ID/roundness at z=20/140/260; roundness ≤0.02; air/3-point bore-gauge record |

가공 route: rough deep drill → stress relieve → semi-finish ream/hone leaving 0.05–0.08 → port/flange machine → gas nitride → final hone → ID/roundness/Ra/case-depth 검사.

Matched drawing-limit diametral clearance는 0.28–0.32, radial clearance는 0.14–0.16이다. Screw OD와 barrel ID를 세 station에서 기록하고 이 범위 안에서 pair한다.

## 1.3. Supplier deliverables

- Material certificate, QT hardness, nitride surface hardness/effective case depth
- Screw OD/TIR/concentricity/straightness, barrel ID/roundness/straightness report
- Ra trace, actual matched clearance table, STEP 기준 DFM deviation list
- 공차 변경 제안은 발주 전 서면 승인; silent substitution 금지

## 1.4. Process coupon release

먼저 `EX-CPN-SCR` L48.00 ±0.05, three RH pitches 16.00 ±0.03, feed-zone OD/root/land와 `EX-CPN-BAR` L60.00 ±0.05, OD34.00 ±0.05, final ID16.20 +0.02/0만 견적·가공한다. Ends는 axis에 0.03 이내 수직이다. Coupon은 동일 material/route/heat treatment로 만들고 OD/ID, pitch, land, Ra, hardness/case depth를 검사한다. Coupon 불합격 또는 질문 미해결이면 full part는 계속 HOLD다.